Facial recognition is real life. And real life is difficult, says NEC

Joseph Brookes 2017-10-26

The challenge confronting companies in the facial recognition market is the ability to operate at speed and scale in real world environments.

“Life isn’t like an e-gate. We don’t live in an e-gate,” said Chris de Silva, head of NEC global face recognition solutions.

Speaking at the company’s NEC Advanced Recognition Systems Experience event in Torquay this week, de Silva was referring to the ideal camera and lighting conditions that allow facial recognition in customs e-gates to work.

“Real life is not easy. Real life is difficult,” he added.

Facial recognition is at the mercy of its hardware and the environment it operates in. Surveillance cameras aren’t usually installed with biometrics in mind, de Silva said. Often they are positioned high with wide angle lenses and poor lighting conditions.

NEC claims their technology sets it apart from competitors because it is “really good at handling those difficult scenarios” and can be utilised on much of the existing surveillance infrastructure.

The Japanese company says it has released “the fastest and most accurate” facial recognition technology that it hopes will place it at the front of the growing biometric industry. NEC claimed their technology led the industry by a good margin but still faced the challenge of how facial recognition is perceived.

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“We are the fastest and most accurate,” de Silva said, lamenting the way Hollywood films had given facial recognition a negative big brother reputation.

“[A] misconception with this stuff is that all faces passing a camera are being sucked up somewhere in the cloud to be used for some strange purpose. That’s not the way it runs,” de Silva told Which-50.

Instead it references images against existing databases to determine a match. A process that happens incredibly quickly and is extremely accurate, de Silva said.

While the technology hasn’t quite reached “James Bond” levels that put a lot of people off it, de Silva insisted there “was a lot that can be done with facial recognition.”

“Some of the face recognition we do is very much leading edge. It’s the most complex type of face recognition,” he said.

That doesn't include real time nationwide surveillance — good news for those who do hold the big brother concern. The realtime application of the software usually involves databases numbering in the tens of thousands. Any more and it begins producing too many false alarms, de Silva said.

"You can do it, you can try to do it, it's just not going to give you much useful information."

"We tend to keep real-time databases low — I think the biggest one that exists in a project is 250,000 people in a database, and that's a bit big, we don't really like it that big, but we allowed that one to happen."

De Silva also stressed the technology had more applications than surveillance and watchlist referencing wasn't inherently negative, pointing to an example of quickly finding lost persons.

"There are many possibilities here in addition to an overt, negative watchlist because people are doing some bad things."

The real time analysis usually leads to one of two scenarios. Referencing the face to an existing database or watch list, or triggering a personalised experience. It presents new analytics opportunities for advertising in addition to the safety and surveillance capabilities, according to NEC Australia's general manager smart systems, Paul Howie.

"There seems to be an opportunity to change the way advertising is being consumed. It's very time sliced at the moment. There is an opportunity with age and gender to break it up on different factors," he said. One example demonstrated was the ability to capture age and gender data on the fly and trigger appropriate advertising.

The Algorithm

The company has been developing facial recognition technology since the late eighties and utilises research labs from around the world that "work as one," according to de Silva. The "engine" of the technology is their heavily guarded algorithm. Even de Silva is not privy to the exact details of how it works. The only people who know that is a few engineers in Japan.

"We are vey cautious with the system," de Silva explained.

"Ultimately nobody knows, other than the core engine designers back in Japan, how that engine works. That's actually the key to it."

Unlike some of their competitors, partners do not have full access to the algorithm. Instead they provide the software as a service which includes systems maintenance, de Silva said.

Following the event de Silva reaffirmed his confidence in the technology, telling Which-50 he had no doubts NEC was the worldwide leader in facial recognition technology, and there was a fair margin to the next competitor. He pointed to 3rd party research conducted by the US government that consistently ranked NEC as leaders.

"The last four times NEC has come number one in that… The design objectives have been to be the fastest, the most accurate and, probably more importantly, able to handle face recognition in really challenging situations."

How it works

According to Paul Howie, NEC Australia general manager smart systems, the software analyses facial landmarks from still images or videos for identifiers to create a “facial signature”. The signature is then cross referenced to an existing database to determine a match.

While the software only uses 2D images, a 3D image is constructed to improve the process.

“We create a 3D representation of a person’s head to increase the range of conditions and angles under which a face can be captured, say for instance if you head is tilted to the side or you are not looking directly at a camera,” Howie said.

In the interests of security, NEC wouldn’t go in to more detail about specifics and the facial landmarks are “only the start of it,” de Silva said.

“We don’t publish exactly what it does and we don’t even share internally exactly what it does.”